EXPERIMENT 7

Network Discovery and Audit with Nmap

Task T:15

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1.Aim

To discover live hosts on an authorized local lab subnet, scan for open TCP ports, identify available services and possible operating-system information with Nmap, and prepare a short security audit report.

2. Course outcome and module

CO4 — Syllabus Program 7: Discover live hosts and open ports on the local subnet, extend the scan to service and OS detection, and write a short security audit report.

3. Tools and versions

Tool / platform	Version used	Purpose / notes
Nmap	7.99 (reported by terminal)	Host discovery, TCP port scan, service/version detection and OS fingerprinting.
Kali Linux	Version not recorded in supplied output	Operating system used for the practical; Kali interface address was 10.0.2.15/24.
Virtual network	NAT-style network indicated by scan fingerprints	10.0.2.2 appears likely to be a virtual gateway; confirm the instructor-assigned target before treating it as the lab host.

4. Program or commands

Commands used in the Kali terminal. The subnet and target must belong to the authorized lab. The first two scans below were run against the virtual network and 10.0.2.2, respectively.

```
ip -4 addr
sudo nmap -sn 10.0.2.0/24
sudo nmap -sS -p 1-1000 10.0.2.2
sudo nmap -sV -O 10.0.2.2 -oN audit.txt
```

Option summary

Option	Meaning
-sn	Host discovery only; skip the port scan.
-sS	TCP SYN scan; run with sudo for raw-packet privileges.
-p 1-1000	Scan TCP ports 1 through 1000.
-sV	Probe open ports to identify service and version information.
-O	Attempt operating-system detection; sudo is generally needed for reliable results.

-oN audit.txt	Save normal-format output to audit.txt in the current directory.

5. How to run (Kali Linux)

1. Open a Kali terminal. Check the interface address and prefix:

```
┌──(kali㉿kali)-[~]
└─$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 85708sec preferred_lft 85708sec
```

2. Discover hosts on the authorized lab subnet:

```
sudo nmap -sn 10.0.2.0/24
```

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-23 21:20 -0400
Nmap scan report for 192.168.1.0
Host is up (0.18s latency).
Nmap scan report for 192.168.1.1
Nmap scan report for 192.168.1.254
Host is up (0.18s latency).
Nmap scan report for 192.168.1.255
Host is up (0.18s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 6.08 seconds
```

3. Select the instructor-provided or otherwise authorized lab device. Scan its first 1,000 TCP ports:

```
sudo nmap -sS -p 1-1000 <authorized-device-IP>
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.1.25
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-23 21:21 -0400
Nmap scan report for 192.168.1.25
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.1.25 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.64 seconds
```

4. Probe services, attempt OS detection, and save a report:

```
sudo nmap -sV -O <authorized-device-IP> -oN audit.txt
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV -O 192.168.1.25 -oN audit.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-23 21:21 -0400
Nmap scan report for 192.168.1.25
Host is up (0.00088s latency).
All 1000 scanned ports on 192.168.1.25 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows S
erver 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Net
work Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo P
LC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.08 seconds
```

5. Read the saved report with:

```
┌──(kali㉿kali)-[~]
└─$ cat audit.txt
# Nmap 7.99 scan initiated Fri Oct 23 21:21:54 2026 as: /usr/lib/nmap/nmap -sV -O -oN audit.txt 192.168.1.25
Nmap scan report for 192.168.1.25
Host is up (0.00088s latency).
All 1000 scanned ports on 192.168.1.25 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows S
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work Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo P
LC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Fri Oct 23 21:22:01 2026 -- 1 IP address (1 host up) scanned in 7.08 seconds
```

6. Sample output and screenshots

The following results are transcribed from the terminal output supplied for this report. They are presented as text because no qualifying screenshot was supplied.

Figure T15-1. Kali interface and live-host discovery

```
Kali interface: 10.0.2.15/24
Live hosts reported: 10.0.2.2, 10.0.2.3, 10.0.2.15
```

```
Host is up (0.18s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 6.08 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 10.0.2.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-23 21:37 -0400
Nmap scan report for 10.0.2.2
Host is up (0.0010s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT    STATE  SERVICE
135/tcp open   msrpc
137/tcp closed netbios-ns
445/tcp open   microsoft-ds
843/tcp open   unknown
MAC Address: 52:55:0A:00:02:02 (Unknown)
```

```
PORT    STATE  SERVICE
135/tcp open   msrpc
137/tcp closed netbios-ns
445/tcp open   microsoft-ds
843/tcp open   unknown
MAC Address: 52:55:0A:00:02:02 (Unknown)
```

7. Result

Host discovery reported three addresses as up on 10.0.2.0/24: 10.0.2.2, 10.0.2.3, and Kali at 10.0.2.15. The scan of 10.0.2.2 found the open TCP ports listed below. The output suggests Windows-related services, but

Nmap could not identify the operating system reliably. Its fingerprint also suggested a virtual network gateway, so the target identity should be confirmed before the findings are attributed to a lab computer.

Host	Open TCP port	Service / version reported	Finding and recommendation
10.0.2.2	135	msrpc; Microsoft Windows RPC	RPC is reachable. Confirm this is an intended lab service and restrict access to trusted lab hosts.
10.0.2.2	445	microsoft-ds; version uncertain	SMB is reachable. Confirm the service is required and limit access; no vulnerability was tested.
10.0.2.2	843	unknown	Service was not identified. Confirm its owner and purpose; disable or restrict it if unnecessary.
10.0.2.2	5357	HTTPAPI 2.0 (Microsoft)	HTTP service detected. Verify that it is expected and limit exposure to the lab.
10.0.2.2	9080	unidentified; HTTP 301 redirect	Nmap could not identify the service. Confirm its purpose before drawing conclusions.

137/tcp was closed. Other scanned ports were reported as filtered (no response). A filtered result does not establish that a service is open or vulnerable. The Nmap output alone does not prove a security weakness.

8. How this version differs from other versions of the same experiment

Comparison	This practical: T15	Related version: T16 (course map)
Interface	Nmap command line in Kali.	Zenmap graphical interface with Nmap.
Recorded version	Nmap 7.99 in the supplied terminal output.	Course map lists Zenmap bundled with Nmap 7.9x.
Workflow	Commands are entered directly; output saved with -oN.	The course map describes using a GUI profile and saving results.
Evidence	Terminal text and Nmap report file.	GUI profile and scan output.

This is a workflow comparison from the supplied course map. No performance comparison between Nmap releases was measured.

9. Viva questions and answers

Q1. What does Nmap -sn do? A. It performs host discovery without a port scan.

Q2. What does the /24 in 10.0.2.0/24 indicate? A. It is the network prefix; here it covers addresses in the 10.0.2.x subnet.

Q3. What is the purpose of -sV? A. It probes open ports to identify their services and, when possible, versions.

Q4. Why use sudo with -sS and -O? A. These scan methods generally need elevated privileges for raw network packet access and OS fingerprinting.

Q5. Does “filtered” mean the port is vulnerable? A. No. It means Nmap did not receive a response that let it determine the port state; it is not proof of a vulnerability.

10. Common errors

Message / symptom	Cause	Fix
zsh: parse error near “\n”	The placeholder <device-IP> was entered literally; angle brackets are shell syntax.	Replace it with the real authorized IP, for example 10.0.2.2, without angle brackets.
All scanned ports filtered	The host did not respond to probes, or traffic is filtered.	Confirm the target IP and lab authorization. Record the result; do not claim ports are open.
OSScan results may be unreliable	Nmap lacks enough open/closed port evidence for a dependable fingerprint.	Treat OS guesses as inconclusive; use a known lab host and report the limitation.
Nmap shows many hosts up unexpectedly	Virtual NAT/proxy behavior or scan conditions can affect discovery results.	Check the assigned subnet and network mode with the instructor; identify the intended lab host.
Could not save or locate audit.txt	The file is saved in the terminal’s current directory.	Run pwd to see the folder, then cat audit.txt to read the saved report.

Sources: Course experiment map and chapter-format instructions supplied by the student; Kali/Nmap terminal output supplied by the student.